Jeffrey Walker

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Summary

Controls, robotics, and autonomy engineer with graduate research in nonlinear flight control and hands-on experience across embedded medical devices, mechatronics, SLAM, perception, and state estimation. Builds simulation-to-implementation workflows with MATLAB/Simulink, Python, C/C++, FlightGear, and microcontrollers, with strongest evidence in NASA-derived F-18 control research, hardware robotics prototyping, and validation-focused engineering communication.

Core Skills

Controls & Autonomy: LQR, PID, sliding-mode control, gain scheduling, nonlinear control, robust control, reinforcement learning, controller evaluation, Monte Carlo robustness analysis

Estimation & Navigation: Kalman filtering, EKF-based sensor fusion, IMU/GPS integration, localization workflows, SLAM pipelines, trajectory alignment, drift and convergence analysis

Simulation & Modeling: MATLAB, Simulink, Simscape, Stateflow, FlightGear, MuJoCo, ROS, Gazebo, PDE Toolbox, dynamics modeling, HIL/SIL experimentation

Programming & Embedded: Python, C/C++, MATLAB, Git, OpenCV, PyTorch, Arduino, ESP32, Raspberry Pi, NVIDIA Orin Nano, PLC logic, actuator and sensor integration

Mechanical & Build: SolidWorks, Ansys, ABAQUS, Autodesk Moldflow, GD&T, 3D printing, CNC, lathe, drilling, prototyping

Education

Drexel University — M.S. Mechanical Engineering, Robotics, Systems & Controls focus — 2023–2025
Thesis: *Advanced Flight Control for Nonlinear Aircraft Models (NASA F-18)* — Philadelphia, PA

Kwame Nkrumah University of Science and Technology — B.S. Automotive Engineering, Land and Electric Vehicle Design — 2019–2023
Project: *Volkswagen Transporter redesign and electrification* — Kumasi, Ghana

Experience

Graduate Research Assistant – Aircraft GNC & Autonomy — *Drexel University, Philadelphia, PA* — Jun 2024 – Present

- Modeled and analyzed a 12-state nonlinear NASA-derived F-18 aircraft system across multiple operating regimes for advanced flight-control research.
- Designed, implemented, and evaluated classical and learning-based controllers including LQR, sliding-mode control, robust-control variants, and reinforcement learning agents.
- Benchmarked controller behavior under disturbance, sensor noise, delays, and actuator saturation, with evaluated runs showing up to **61% reduction in peak state excursions** across primary stability-related states.
- Built validation workflows in MATLAB/Simulink and FlightGear to study closed-loop behavior, visual proof, and simulation-to-deployment considerations.
- Authored thesis-level documentation covering architecture, assumptions, controller comparisons, and evaluation results.

Controls and Embedded Systems Intern – Medical Devices — *T.I. Inc, Pennsylvania* — Sep 2024 – Feb 2025

- Developed embedded control and interface logic for neonatal medical-device subsystems using Arduino, ESP32, C/C++, and Python.

- Integrated sensors, actuators, displays, and motor drivers into electromechanical prototypes with real-time response constraints.
- Implemented watchdog behavior, fail-safe logic, and structured test workflows to improve prototype reliability.
- Contributed to verification and validation activities aligned with **IEC 62304** and **ISO 14971** in a regulated engineering environment.

Aerospace Controller Design Researcher *Drexel University – Prof. Bor Ching Chang, Philadelphia, PA* Jul 2024 – Present

- Supported aerospace controls research involving robust controller design with H_2 and H_∞ methods.
- Contributed to modeling, simulation, and controller-development workflows in MATLAB/Simulink for high-precision aerospace applications.
- Helped bridge simulation work toward embedded or hardware-oriented experimentation through Raspberry Pi-based workflows.

Industrial Automation Systems Researcher *Drexel University – Prof. Dela Fuentes, Philadelphia, PA* May 2024 – Present

- Conceptualized closed-loop industrial automation architectures using sensing, actuation, and PLC-oriented logic.
- Built prototype hardware using additive and subtractive manufacturing workflows at the Drexel Innovation Studio.
- Supported system design at the intersection of controls, automation, prototyping, and manufacturability.

Automotive Engineering Workshop Staff / Engineer *KNUST, Kumasi, Ghana* Nov 2022 – Sep 2023

- Supported workshop engineering activities spanning manufacturing, maintenance, prototyping, and vehicle-system troubleshooting.
- Used CNC, lathe, and drilling machines to manufacture or refine automotive components, contributing to a reported **10% reduction in part failure rates**.
- Applied MATLAB-based process and lab optimization methods that reduced test times by **20%**.
- Supported maintenance and diagnostic work on more than **60 vehicles**, strengthening hands-on mechanical and systems intuition.

Selected Projects

Nonlinear F-18 Flight Control *Controls / RL / Simulation* 2024 – 2025

- Built a nonlinear aircraft control workflow around a NASA-derived F-18 model using classical control and reinforcement learning methods.
- Organized the project around open-loop versus closed-loop comparison, FlightGear visualization, and technical interpretation rather than abstract AI claims.

SpiRob Cable-Driven Soft Robot *Embedded Control / Soft Robotics / Mechatronics* 2024

- Developed a spinal-inspired soft robot with cable actuation, stepper-motor control, and modular embedded logic for wave-like motion experiments.
- Combined physical prototyping, embedded actuation, and MuJoCo-oriented modeling to connect hardware work with future autonomy and control development.

Stereo SLAM Pipeline for UAV Workflows *SLAM / Perception / Evaluation* 2025

- Built a stereo SLAM workflow with KITTI-style data recording, pluggable backend structure, and trajectory evaluation.

- Structured the project around pose visualization, alignment, and interpretation rather than treating SLAM as a black-box demo.

Vision Safety Monitoring System *Computer Vision / OpenCV / Monitoring* 2025 – 2026

- Built an OpenCV-based monitoring pipeline with interactive no-go zones, posture labeling, event detection, and alert logging.
- Designed the workflow for live or file-based execution with practical operator interaction and deployment-oriented structure.

Sensor Fusion Navigation *Estimation / EKF / Localization* 2025

- Implemented EKF-based fusion of IMU and GPS signals for ground-vehicle navigation under noisy measurements and intermittent dropout.
- Studied localization behavior across representative trajectories with a focus on robustness and estimation quality.

Leadership, Honors & Academic Work

- **Dean's Fellowship**, Drexel University 2023 – 2025
- **Leroy L. Resser Endowment Scholarship** 2024
- **College of Engineering Student Ambassador**, Drexel University 2024 – 2025
- **Volunteer Mentor and Liaison**, KNUST Outreach Program Aug 2024 – Present
- **Master's Thesis:** *F-18 Aircraft Dynamical Control Using Classical Control Methods and Reinforcement Learning: A Comprehensive Study*, Drexel University, Aug 2025